ACKNOWLEDGMENTS

This work has received funding from the Science Foundation Ireland's Future Digital Challenge programme for the project entitled "Exploring realistic pathways to the decarbonization of buildings in the urban context" under the grant No. 22/NCF/FD/10985.

REFERENCES

Usman Ali, Mohammad Haris Shamsi, Mark Bohacek, Karl Purcell, Cathal Hoare, Eleni Mangina, and James O'Donnell. A data-driven approach for multi-scale gis-based building energy modeling for analysis, planning and support decision making. *Applied Energy*, 279:115834, 2020.

Dara Bahri, Heinrich Jiang, Yi Tay, and Donald Metzler. Scarf: Self-supervised contrastive learning using random feature corruption. In *International Conference on Learning Representations*, 2021.

Monica Billio, Michele Costola, Loriana Pelizzon, and Max Riedel. Buildings' energy efficiency and the probability of mortgage default: The dutch case. *The Journal of Real Estate Finance and Economics*, 65(3):419–450, 2022.

Bryan Coyne and Eleanor Denny. Mind the energy performance gap: testing the accuracy of building energy performance certificates in ireland. *Energy efficiency*, 14(6):57, 2021.

Lukas Dahlström, Tor Broström, and Joakim Widén. Advancing urban building energy modelling through new model components and applications: A review. *Energy and Buildings*, 266, 2022.

Hilary Egan, Clement Fouquet, and Chioke Harris. Machine learning for advanced building construction. In *ICLR 2023 Workshop on Tackling Climate Change with Machine Learning*, 2023.

Emerald Energy and Esbensen Consulting Engineers. Energy efficiency regulations for new dwellings and options for improvement.

A APPENDIX

A.1 DATA PREPROCESSING & FEATURE EXTRACTION

The data preprocessing consists of three steps. The first step is data cleaning. In this step, we use the interquartile range (IQR) technique to identify and address outliers. We then group the data by building type. For numerical features, missing values are imputed with the mean value specific to each building type. For categorical features, we utilized the most frequent value in each category for fill-ins. In the second step, we selected 40 features from the total 211 features in the original dataset. The selection is based on the feature importance analysis with decision tree algorithm, which identified the features with the most significant impact on energy ratings. During this process, we excluded compound features that can be derived by the actual energy ratings, such as total primary energy use for the dwelling and the dwelling's carbon dioxide emissions (known as CO2 emissions). These 40 features are categorised into distinct groups, each describing different aspects of a building. These categories include building envelope features (e.g., the area of the wall, roof, and door), building fabric features (e.g., the U-values for the wall, roof, and door), characteristics of the heating system (e.g., the main heating system efficiency), hot water-related attributes (e.g., water storage volume), and spatial features (e.g., the county code). Lastly, in the third step, we applied standard scalers to standardize the numerical features and one-hot encoders to transform categorical features.

A.2 DATASET

We use the Energy Performance Certificates (EPC) dataset [2] in this study, collected by the Sustainable Energy Authority of Ireland. This dataset contains 112,528 building assessment records. There are 15 different BER levels in total, ranging from A1 as the highest level to G as the lowest level.

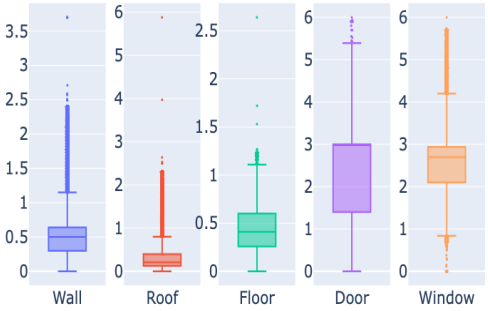

Figure 3: Box-plots of U-Values

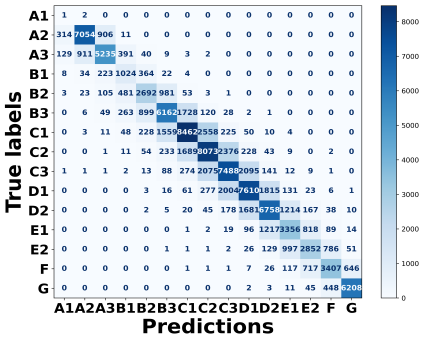

Figure 4: Confusion Matrix for MLP Predictions

A.3 Experiments Configuration

In our experiments, the SCARF model was trained for 15 epochs, with a batch size of 16 and a learning rate of 0.001. The model used in the experiments consists of three MLP blocks, each with a linear layer followed by ReLU activation. We set the dimension of the model's encoder to be 32. The proportion of randomly selected feature corruption is set to be 30%. We split the dataset into train, validation, and test sets by the ratio 80%, 10%, and 10%.

[2]https://ndber.seai.ie/BERResearchTool/ber/search.aspx